

LAB 7: Configuration of Switch, VLAN Configuration and Inter-VLAN Routing

Objectives

- To study the basic configuration and operation of a network switch.
- To understand the concept and purpose of VLANs in a network.
- To configure VLANs and assign switch ports to different VLANs.
- To implement and verify inter-VLAN routing between different VLANs.

Software and Hardware Requirements

- Cisco Packet Tracer
- Windows Operating System
- Computer/Laptop
- Keyboard and Mouse

Theory

Network Switch

A network switch is a Layer-2 networking device that connects multiple devices in a Local Area Network (LAN). It operates at the Data Link Layer of the OSI model and forwards data frames based on the MAC addresses of devices. Unlike a hub, a switch creates separate collision domains for each connected device, which reduces collisions and improves network performance. Managed switches provide additional features such as VLAN configuration, port security, and trunking. Switches are essential for efficiently managing network traffic and ensuring reliable communication between devices within the same network.

Virtual Local Area Network (VLAN)

A VLAN is a logical grouping of devices within a LAN, allowing devices to communicate as if they are on the same network, regardless of their physical location. VLANs divide a

network into multiple broadcast domains, which helps reduce unnecessary traffic, enhances security, and simplifies network management. For example, different departments like HR, IT, and Finance can be placed in separate VLANs for better organization and privacy. There are several types of VLANs, including default VLANs, data VLANs for user traffic, management VLANs for switch administration, and native VLANs for untagged traffic on trunk links.

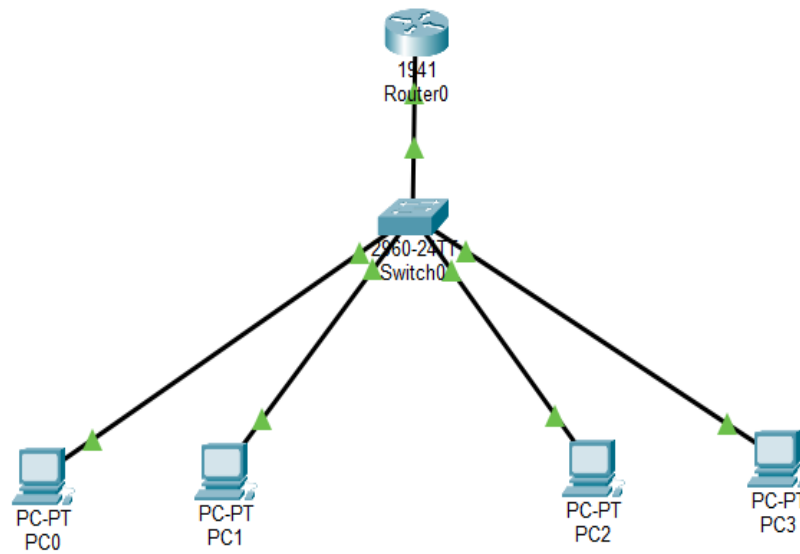
VLAN Configuration

Configuring VLANs on a switch involves creating VLANs with specific IDs, assigning switch ports as access ports for end devices, and configuring trunk ports to carry traffic from multiple VLANs between switches. Access ports belong to a single VLAN and are used for devices like computers and printers. Trunk ports, on the other hand, connect switches or routers and carry traffic for multiple VLANs using IEEE 802.1Q tagging. VLAN configuration allows better traffic segmentation, reduces broadcast domains, and ensures that only intended devices communicate within a VLAN.

Inter-VLAN Routing

By default, devices in different VLANs cannot communicate because each VLAN is a separate broadcast domain. Inter-VLAN routing enables communication between VLANs using a router or a Layer-3 switch. In the router-on-a-stick method, a single router interface is divided into multiple sub-interfaces, each assigned to a VLAN with its own IP address. A Layer-3 switch can also perform routing internally using Switch Virtual Interfaces (SVIs), which is faster and more efficient than using a router. Inter-VLAN routing improves network organization, allows controlled communication between departments, and enhances security by managing traffic flow between VLANs.

NETWORK TOPOLOGY



CONFIGURATION

VLAN and IP Addressing Scheme

VLAN ID	VLAN Name	Network Address	Gateway
10	Computer	192.168.10.0/24	192.168.10.1
20	Electronics	192.168.20.0/24	192.168.20.1

Device	IPv4 Address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3	192.168.20.3	255.255.255.0	192.168.20.1

COMMANDS

Switch Configuration Commands

1. Create VLANs

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Computer
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# name Electronics
```

```
Switch(config-vlan)# exit
```

2. Assign Ports to VLANs

```
Switch(config)# interface range fa0/1-2
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 10
```

```
Switch(config-if-range)# exit
```

```
Switch(config)# interface range fa0/3-4
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 20
```

```
Switch(config-if-range)# exit
```

3. Configure Trunk Port (Switch to Router)

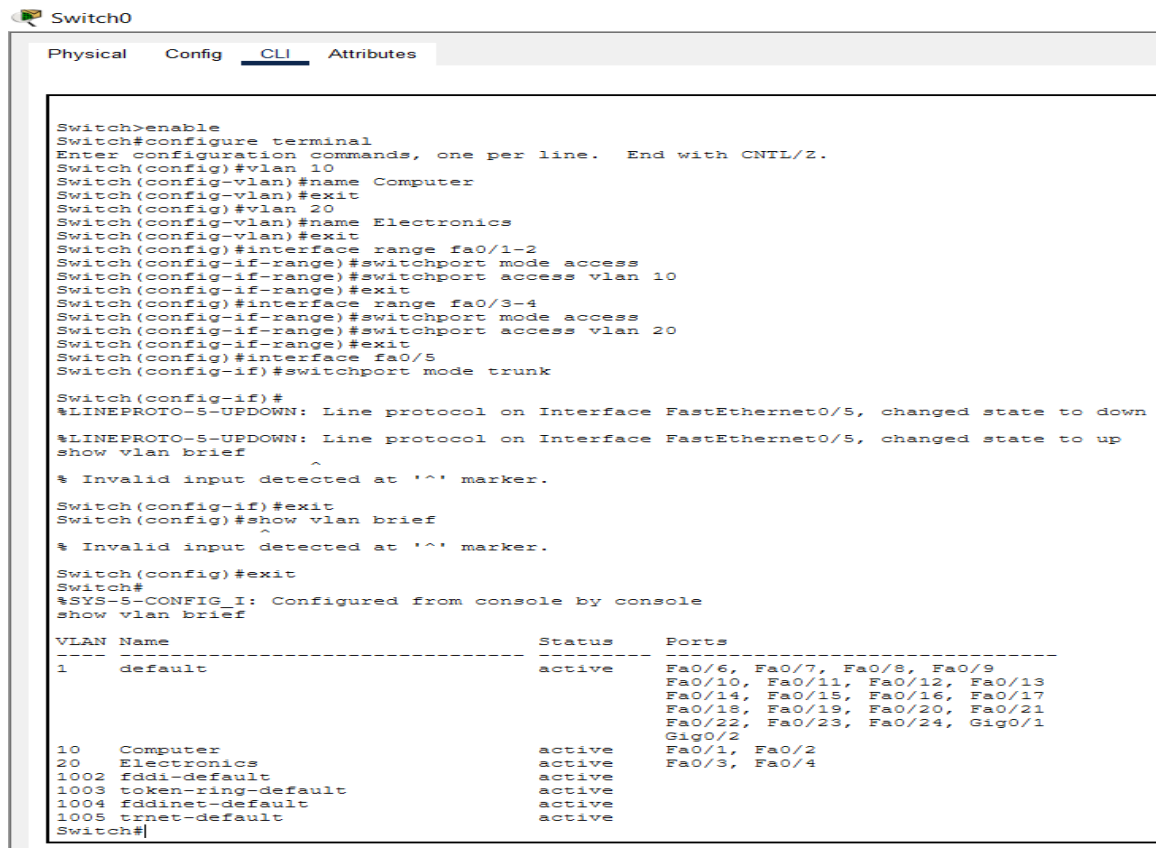
```
Switch(config)# interface fa0/5
```

```
Switch(config-if)# switchport mode trunk
```

```
Switch(config-if)# exit
```

Verification Switch

Switch# show vlan brief



```
Switch0
Physical Config CLI Attributes

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Computer
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Electronics
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface fa0/5
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
show vlan brief
^
% Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#show vlan brief
^
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#
%SYS-5-CONFIG I: Configured from console by console
show vlan brief

VLAN Name                Status    Ports
----
1    default                active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Fa0/24, Giga0/1
                                           Giga0/2
10   Computer                active    Fa0/1, Fa0/2
20   Electronics             active    Fa0/3, Fa0/4
1002 fddi-default            active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
Switch#
```

Router Configuration Commands

1. Configure Sub-Interfaces for VLANs

Router(config)# interface GigabitEthernet 0/0.10

Router(config-subif)# encapsulation dot1Q 10

Router(config-subif)# ip address 192.168.10.1 255.255.255.0

Router(config-subif)# exit

Router(config)# interface GigabitEthernet 0/0.20

Router(config-subif)# encapsulation dot1Q 20

Router(config-subif)# ip address 192.168.20.1 255.255.255.0

Router(config-subif)# exit

```
Router0
to up
encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface GigabitEthernet 0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state
to up
encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface GigabitEthernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface GigabitEthernet 0/0.10
Router(config-subif)#exit
Router(config)#interface GigabitEthernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface GigabitEthernet 0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

PING OUTPUT

Ping PC0-PC1

```
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Ping PC1-PC3

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.2: bytes=32 time=12ms TTL=127
Reply from 192.168.20.2: bytes=32 time=12ms TTL=127
Reply from 192.168.20.2: bytes=32 time=13ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 13ms, Average = 12ms

C:\>
```

RESULTS

The network topology was successfully implemented, achieving logical segmentation and enabling VLAN communication. VLAN verification using the `show vlan brief` command confirmed that VLAN 10 (Computer) with ports Fa0/1–Fa0/2 and VLAN 20 (Electronics) with ports Fa0/3–Fa0/4 were active. ICMP ping tests showed successful intra-VLAN communication between PC0 and PC1 in VLAN 10 with 0% packet loss. Inter-VLAN communication was verified as PC1 (VLAN 10) successfully pinged PC3 (VLAN 20) after initial ARP resolution, confirming that the Router-on-a-Stick configuration worked correctly.

DISCUSSION

The lab shows that a Layer-2 switch creates separate collision domains, improving network performance. VLANs logically segmented the network into distinct broadcast domains, reducing unnecessary traffic and isolating departments unless routing is configured. Inter-VLAN routing was achieved through a single router interface with sub-interfaces for VLAN 10 and 20 using IEEE 802.1Q encapsulation. The trunk port on the switch carried traffic for both VLANs. This configuration also demonstrates enhanced security and management, as each VLAN has its own default gateway, controlling traffic flow between departments.

CONCLUSION

The lab objectives were achieved successfully. It demonstrated the operation of managed Layer-2 switches, VLAN creation and port assignment, and trunk port configuration. Inter-VLAN communication via Router-on-a-Stick showed that logically separated VLANs can communicate efficiently and securely through a Layer-3 device. Overall, VLANs improve network organization, performance, and security while enabling controlled communication between departments.